ENVS 193DS Final

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Github repository

```
# reading in packages
library(tidyverse)
library(janitor)
library(here)
library(ggeffects)
library(MuMIn)
library(dplyr)
library(rstatix)
# creating new object from csv in `data` folder
# using "here" to identify location of file
nests <- read_csv(here("data", "nest_data_final.csv"))
# creating new value for removed variables in locations column
loc_removed <- c("Cafe", "Amtrak",
                 "Boyfriend's house", "UCen",
                 "Arbor", "SRB")
# creating new object from personal data
# # using "here" to identify location of file
personal <- read_csv(here("data",
                          "personal.csv")) |>
  # cleaning columnn names
  clean_names() |>
  # removing loctations with fewer than 5 observations
  filter(!(location %in% loc_removed))
```

Problem 1: Research writing

(a)

In part 1 they used a linear model.
In part 2 they used a Pearson's test.

(b)

We found that the distance to the water's edge tends to predict the probability that American Avocets would use a habitat. We also found a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and the type of habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

(c)

One additional variable that could influence the probability of American Avocet microhabitat use is the type of habitat, which is a categorical variable. They may have preferred food sources or shelter in some habitats and not others, which could influence the probability of microhabitat use.

Problem 2: Data analysis

(a)

```r
# creating new object from nests dataframe
nests_clean <- nests |>
  # creating new column with full names of ants
  mutate(species_name = recode_values(
    ant_species,
    "RRB" ~ "C. mimosae",
    "AB" ~ "C. sjostedti",
    "BBR" ~ "C. nigriceps"
  )) |>
  # reordering species levels
  mutate(species_name =
           as_factor(species_name),
         species_name =
           fct_relevel(species_name,
                       "C. sjostedti",
                       "C. mimosae",
                       "C. nigriceps"
           )) |>
  # selecting columns of interest (species name, case_control, and height_cm)
  select(species_name, case_control, height_cm) |>
  # dropping N/A values from species name column
  drop_na(species_name)
# checking structure of data frame
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ species_name: Factor w/ 3 levels "C. sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
```

```r
# displaying from nests_clean
slice_sample(
  nests_clean,
  # displaying 10 rows
  n = 10)
```

```
# A tibble: 10 x 3
   species_name case_control height_cm
   <fct>               <dbl>     <dbl>
 1 C. mimosae              1      525
 2 C. mimosae              0      427
 3 C. nigriceps            0       54
 4 C. mimosae              0      329
 5 C. mimosae              0      300
 6 C. nigriceps            0      105
 7 C. mimosae              1      201.
 8 C. mimosae              0      212.
 9 C. mimosae              1      212.
10 C. mimosae              0      210
```

(b)

In this data set, the 0s and 1s represent whether a tree contained a nest (1) or was an "available" tree for comparison (0).

(c)

Tree height is a continuous variable. I would expect that as height increases, the probability of nesting would also increase. This is supported by the article, which says that for every 1m increase in height, the odds that birds nested in a tree increased by 20% (Results section, paragraph 2).

Acacia ant species are categorical variables. I would expect that in trees with C. nigriceps and C. mimosae, there would be a higher probability of nest occurrence. Based on the article, it seems that these ants are the most aggressive. This seems to increase probability of nests because these ants may help reduce the risk of nest predation (Discussion section of the article in paragraph 1).

(d)

Model number	Tree height	Ant species	Interaction term
0	X	X	No
1	X	X	Yes

(e)

```
# model 0: additive model
model1 <- glm(
  # formula: species and height
  case_control ~ species_name + height_cm,
  # from clean data frame
  data = nests_clean,
  family = binomial(link = "logit")
)
# model 2: interactive model
model2 <- glm(
  # formula: height interacting with species
  case_control ~ height_cm * species_name,
  # from clean data frame
  data = nests_clean,
 family = binomial(link = "logit"))
```

(f)

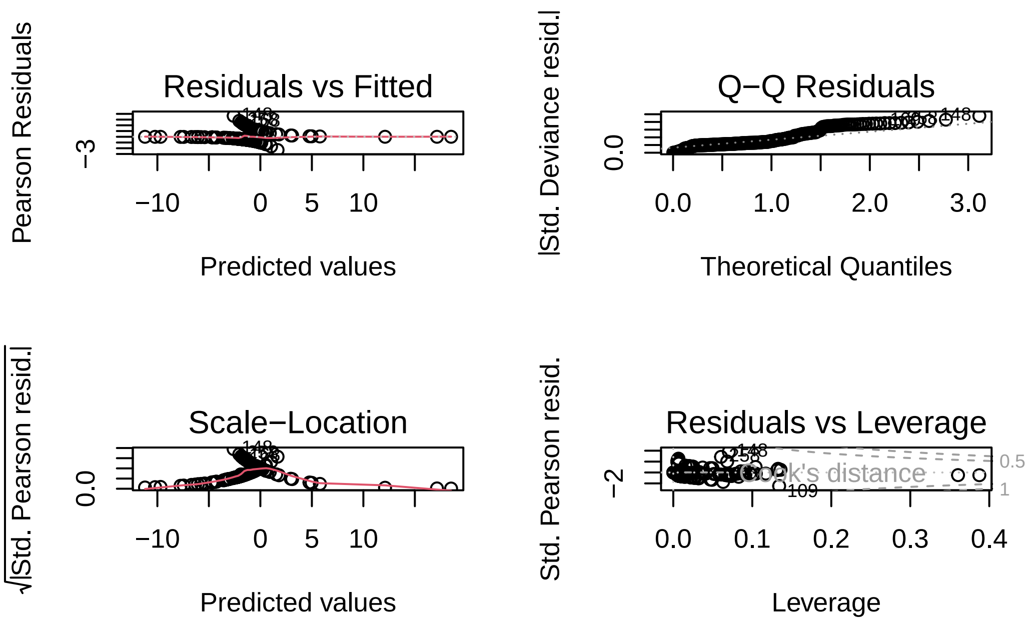
```
# calculating AIC
AICc(
  model1,
  model2
) |>
  # arranging by AIC
  arrange(AICc)
```

```
       df     AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) is the one that uses the interaction ant species and tree height to predict nest occurence.

(g)

```
# displaying plots in 2x2 grid
par(mfrow = c(2, 2))
#
plot(model2)
```




```
summary(model2)
```

```
Call:
glm(formula = case_control ~ height_cm * species_name, family = binomial(link = "logit"),
    data = nests_clean)

Coefficients:
                                         Estimate Std. Error z value Pr(>|z|)
(Intercept)                             -3.693410   2.517395  -1.467  0.14233
```

```
height_cm                              0.003895   0.008237   0.473  0.63634
species_nameC. mimosae                 0.848334   2.554617   0.332  0.73983
species_nameC. nigriceps             -12.279738   5.946723  -2.065  0.03893 *
height_cm:species_nameC. mimosae       0.002597   0.008386   0.310  0.75679
height_cm:species_nameC. nigriceps     0.084541   0.031961   2.645  0.00817 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

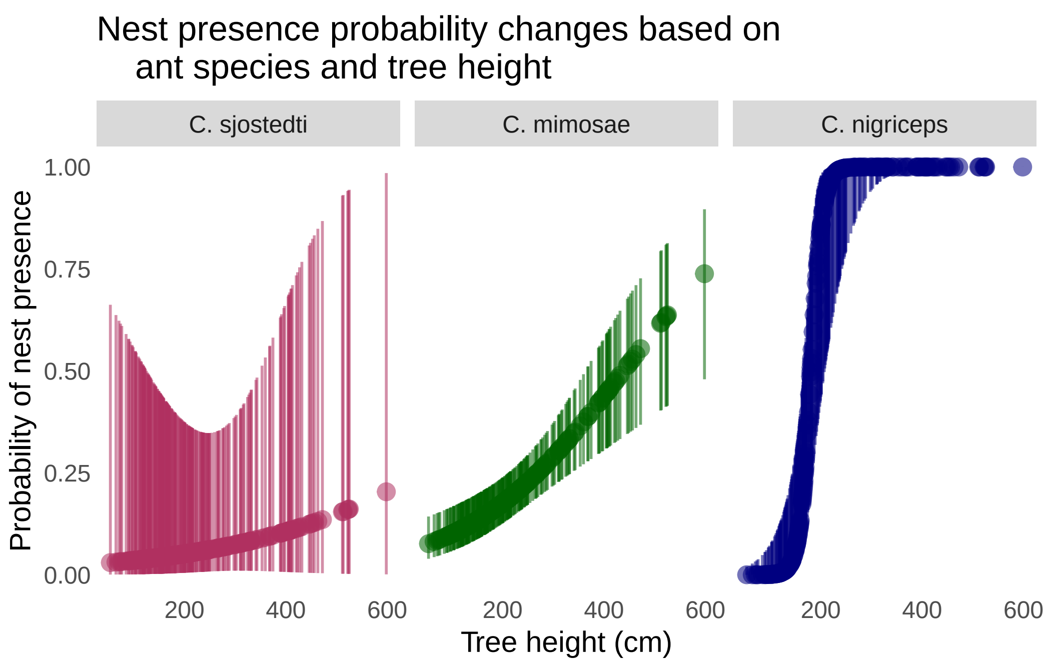
(h)

```r
preds <- ggpredict(
  model2,
  terms = c("height_cm [all]", "species_name"))
```

```r
ggplot(data = nests_clean,
       mapping = aes(x = x,
                     y = case_control,
                     color = group)) +
  geom_pointrange(data = preds,
                  mapping = aes(x = x,
                                y = predicted,
                                ymin = conf.low,
                                ymax = conf.high,
                                alpha = 0.1)) +

  scale_color_manual(values = c("maroon", "darkgreen", "navy")) +
  facet_wrap(~group,
             scales = "free_x") +
  labs(
    x = "Tree height (cm)",
    y = "Probability of nest presence",
    title = "Nest presence probability changes based on
    ant species and tree height"
```

```
  ) +
  theme(legend.position = "none",
        panel.background = element_blank(),
    panel.grid = element_blank(),
    axis.ticks = element_blank())
```



(i)

Figure 1: Impact of ant species and tree height on birds nest occurrence. Each panel represents the presence of a different species of ant. Maroon shows the data for C. sjostedti, green shows the data for C. mimosae, and navy shows the data for C. nigriceps. The points represent the probability of the presence of a birds nest with 1 being a birds nest and a 0 being no nest. These predictions are based on both the species and the height of the tree. The lines extending off the points represent a 95% confidence interval.

(j)

```
# calculating the presence prediction at height = 600 cm
ggpredict(
  model2, # model object
```

```
  terms = c("height_cm [600]",
            "species_name [C. nigriceps]")
)
```

```
# Predicted probabilities of case_control

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

(k)

I found that the simplest model that represented the most variation in nest presence included tree height (cm) and ant species as interacting variables. Specifically, the interaction between C. nigriceps and height was a significant predictor of nest presence ($\beta = 0.084$ z = 2.645, p = 0.008, $\alpha = 0.001$). This indicates a strong relationship between tree height, presence of C. nigriceps ants, the occurrence of birds nests. The predicted probability of nest site occupancy at 600 cm with these variables is 1.00 (95% Ci 1.00, 1.00). This makes sense with the paper, as these ants are aggressive, which can protect the birds and their nests from predators.

Problem 3: Affective and exploratory visualizations

(a)

The exploratory visualizations I made in Homework 2 are very different from the affective visualization I made for Homework 3. The exploratory visualizations were made to get information across in a simple and understandable way. The affective visualization contains data, but that is not the only goal. It is more about being visually interesting with the explanation of the data being represented being on another sheet of paper.

They are all similar in that they come from the same dataset. The affective visualization is also similar to the boxplot from the exploratory visualization. They bot display the productivty at different locations.

In the affective visualization, the mean productivity levels are displayed, which show a similar pattern to the boxplot, which shows the median productivity among other things. The productivity based on the time of start does not share any patterns with the other two that I can determine.

In week 9, my group thought my idea was good. However, some members suggested that I add other variables from my data. I did not end up doing this because I was unsure how to do that in a way that would make sense logically and artistically.

(b)

I will do a Kruskall-Wallis test to compare the median productivity rank at each location. It will tell me if there is a significant difference in my productivity levels in the library compared to at home. Since my predictor variable (location) is two unordered groups, it will work well for this test. My response variable (productivity) is ordinal, which works well for a Kruskall-Wallis test since it compares the ranks of scores and does not assume a normal distribution. Each study session is an independent observation.

(c)

Checking assumptions

1. Independence Each study session is an independent observation.
2. Sample size: each group should have at least 5 observations. Locations that had fewer than 5 observations were removed, leaving "Home" and "Library".

```
personal |>
  count(location, sort = TRUE)
```

```
# A tibble: 2 x 2
  location     n
  <chr>    <int>
1 Home        20
2 Library      7
```

```
dunn_test(
  productivity_rating_1_5 ~ location,
  data = personal,
  p.adjust.method = "bonferroni"
)
```

```
# A tibble: 1 x 9
  .y.               group1 group2    n1    n2 statistic     p p.adj p.adj.signif
* <chr>             <chr>  <chr>  <int> <int>     <dbl> <dbl> <dbl> <chr>
1 productivity_rat~ Home   Libra~    20     7     0.484 0.629 0.629 ns
```

Running analysis

```r
# running kruskal-wallis test
kruskal.test(
  # productivty as a function of location
  productivity_rating_1_5 ~ location,
  # data: personal data
  data = personal
)
```

```
    Kruskal-Wallis rank sum test

data:  productivity_rating_1_5 by location
Kruskal-Wallis chi-squared = 0.23382, df = 1, p-value = 0.6287
```

```r
# starting with personal data
personal |>
  # grouping by location
  group_by(location) |>
  # calculating median productivity for each location
  summarise(median_prod =
            median(productivity_rating_1_5))
```

```
# A tibble: 2 x 2
  location median_prod
  <chr>          <dbl>
1 Home             4.5
2 Library          5
```

Effect size

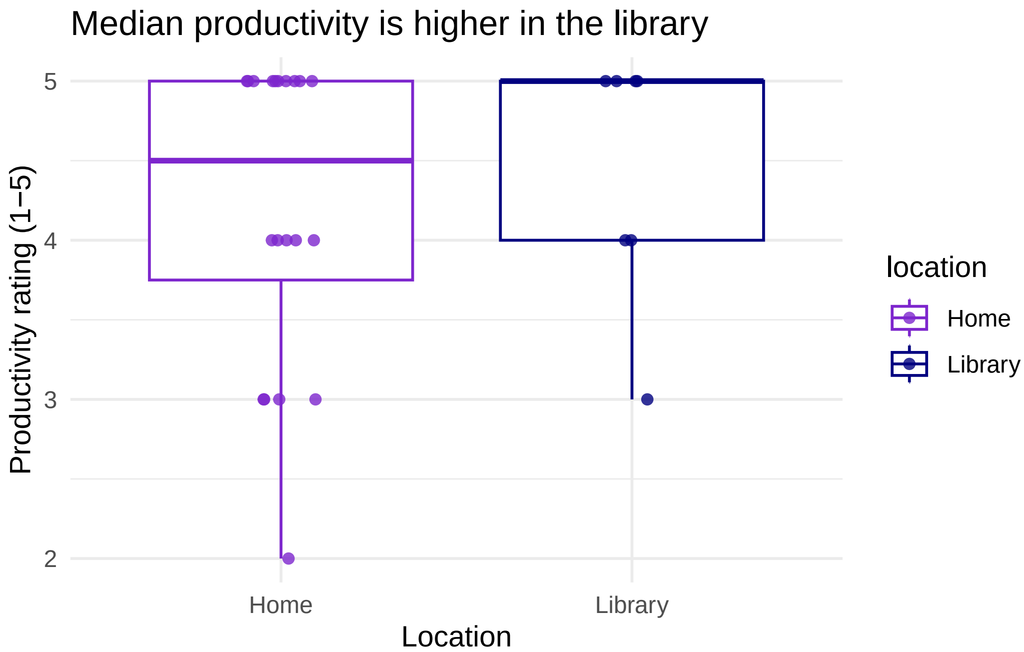
```r
# determining effect size
kruskal_effsize(
  # productivity rating as a function of location
  productivity_rating_1_5 ~ location,
  # data: personal data
  data = personal)
```

```
# A tibble: 1 x 5
```

```
  .y.                           n effsize method  magnitude
* <chr>                     <int>   <dbl> <chr>   <ord>
1 productivity_rating_1_5     27 -0.0306 eta2[H] small
```

(d)

```
# base layer ggplot
# data: personal data
ggplot(data = personal,
       # x-axis: location
       # y-axis: productivity rating
       mapping = aes(x = location,
                     y = productivity_rating_1_5,
                     # color by location
                     color = location)) +
  # first layer: boxplot
  geom_boxplot() +
  # second layer: jitter plot
  # adjusted height, width, opacity of points
  geom_jitter(height = 0,
              width = 0.1,
              alpha = 0.8) +
  # changing colors
  scale_color_manual(values = c("purple3", "navy")) +
  # changing axis titles and adding a title
  labs(x = "Location",
       y = "Productivity rating (1-5)",
       title = "Median productivity is higher in the library") +
  # minimal theme
  theme_minimal()
```



(e)

Figure 1: Comparison of median productivity levels at home and the library. Each boxplot represents a location where I have studied, and each point represents a single study session. The bold line in the boxes represent the median value, and the top and bottom of the boxes represent the 75th and 25th percentiles respectively. The whiskers span the distance of the lowest to the highest 25% of the data.

(f)

I found a small (2 = 0.03) difference in median productivity between the library and my home (Kruskal-Wallis test, $\chi^2(1)= 0.234$, p = 0.629, = 0.05). There was not a significant difference in median productivity between the library (median productivity = 5) and my home (median productivity = 4.5) (Dunn's post hoc test: adjusted p-value= 0.629).